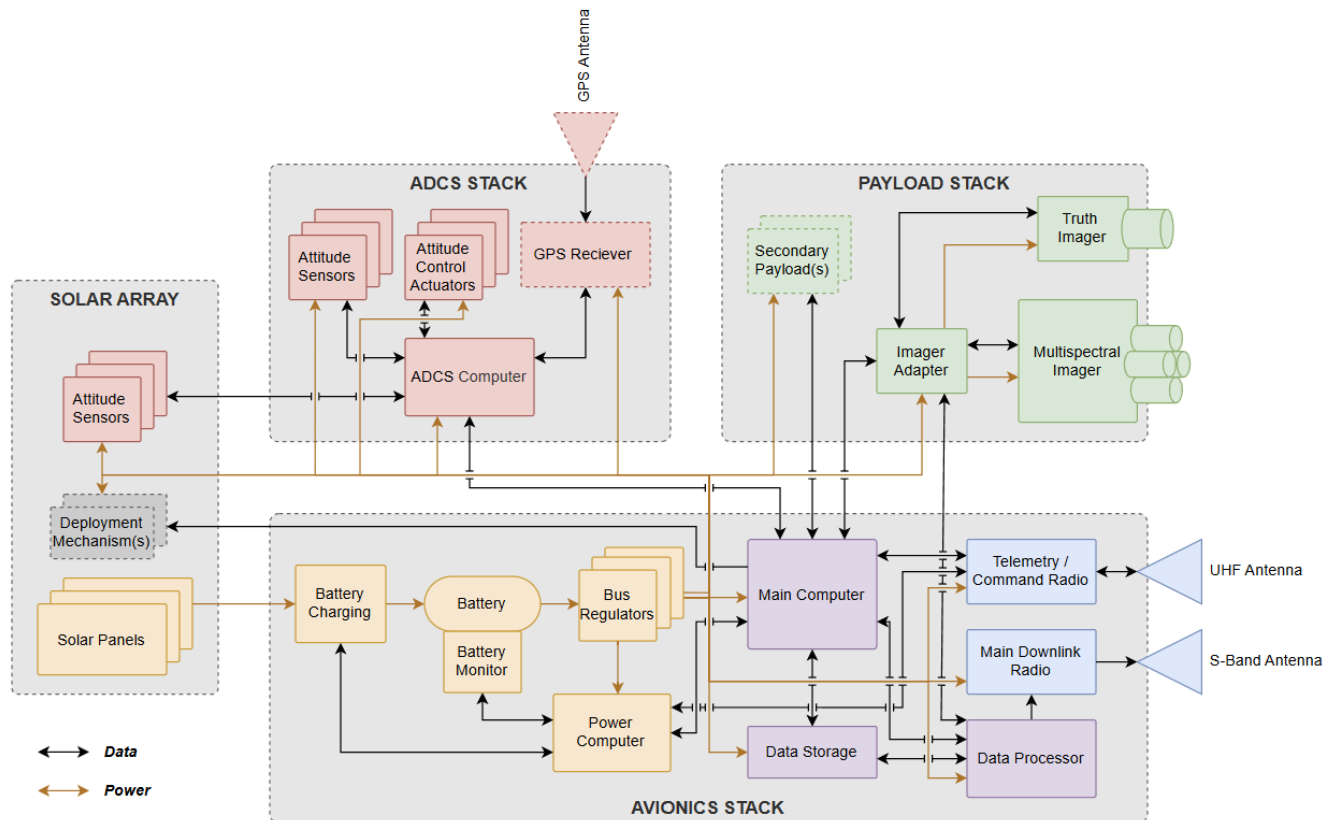


Assignment 5

Caden O'Leary, Elmoataz Abdellah, Matthew Goldsberry, Seunghun Lee, Kristen Griessman

Block Diagrams

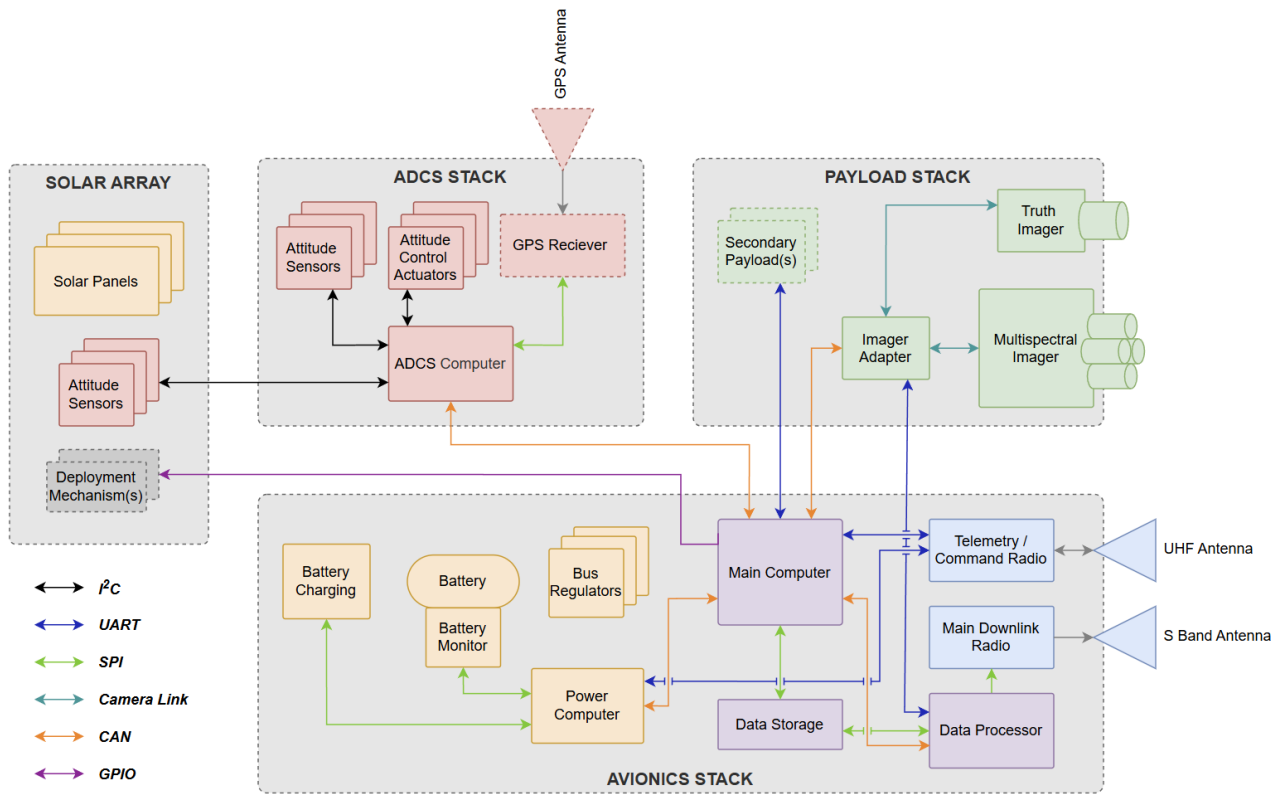
HABSat-1 Data and Power Lines



HABSat-1 Data & Power Lines

Defines the power and dataflow between components in HABSat-1.

HABSat-1 Protocol Lines



HABSat-1 Protocol Lines

Defines the hardware communication protocols between components in HABSat-1.

Component Responsibility Table

ADCS Stack

Component	Responsibility	In	Out	Owner
Attitude Sensors	Provide sensor readings to determine the satellite's orientation and the location of the Sun and Earth.	ADCS Computer	ADCS Computer	ADCS Subsystem
Attitude Control Actuators	Reaction wheels that spin to control satellite orientation.	ADCS Computer	ADCS Computer	ADCS Subsystem
GPS Receiver	Reads antenna data to determine location.	ADCS Computer	ADCS Computer	ADCS Subsystem
GPS Antenna	Receives GPS data.	Sensor Data	GPS Antenna	ADCS Subsystem
ADCS Computer	Reads sensor data and TLEs to determine future passes, orient the satellite to target, orient the solar panels towards the sun, orient the satellite towards the ground station for downlink, and detumble the satellite upon launch.	Attitude Sensor GPS Receiver	Attitude Sensors (Solar Array, ADCS Stack) Attitude Sensor Actuator	ADCS Subsystem Seunghun

Payload Stack

Component	Responsibility	In	Out	Owner
Secondary Payload	TBD. Working to find a partner for this.	Main Computer	Main Computer	Payload Subsystem Caden
Imager Adapter	Controls the truth and multispectral imager.	Main Computer	Main Computer Multispectral Imager Truth Imager	Payload Subsystem Kristen
Multispectral Imager	Captures images matching the color wavelengths of light for algae.	Multispectral Imager	Image Adapter	Payload Subsystem
Truth Imager	Captures the real image of the lake to match multispectral image data.	True images	Image Adapter	Payload Subsystem

Avionics Stack

Component	Responsibility	In	Out	Owner
Main Computer	Orchestrate mission operations based on satellite state by commanding other components to complete tasks.	ACDS Computer Secondary Payload Imager Adapter Telemetry / Command Radio Data Processors Data Storage Power Computer	ACDS Computer Secondary Payload Imager Adapter Telemetry / Command Radio Data Processors Data Storage Power Computer	Matthew Caden Elmoataz Kristen Seunghun
Data Storage	Storage of payload and mission software	Data Processor Main Computer	Data Processor Main Computer	Elmoataz

Data Processor	Load images and perform chlorophyll index conversion on multispectral images.			Matthew
Main Downlink Radio	Downlink of multispectral images	S-Band Antenna	Data Processor	Comms Subsystem Caden
S-Band Antenna	Transmit image data via S-band.	S-band Signals	Main Downlink Radio	Comms Subsystem
Telemetry / Command Radio	Uplink commands from ground station and downlink	UHF Antenna	Main Computer	Comms Subsystem Caden
UHF Antenna	Transmit and receive UHF signals	UHF Signals	Telemetry / Command Radio	Comms Subsystem
Power Computer	Monitors power state and ensure the satellite stays within safe thresholds, while also providing insight to the Main Computer	Battery Charging Battery Monitor Main Computer Telemetry / Command Radio	Main Computer, Battery Charging, Telemetry / Command Radio	Power Subsystem Seunghun
Battery Charging	Regulate charging current from solar panels to safely charge the satellite battery.	Solar Panels, Power Computer	Battery, Power Computer	Power Subsystem
Battery	Store energy generated by solar panels and supply	Battery Charging	Bus Regulators, Battery Monitor	Power Subsystem

	direct power to the power bus.			
Battery Monitor	Measure real time battery voltage, current, and temperature to report state of charge.	Battery	Power Computer	Power Subsystem
Bus Regulators	Step down and stabilize battery voltage to deliver regulated power lines to avionics components.	Battery	Main Computer, Power Computer, Subsystems	Power Subsystem

Solar Array

Component	Responsibility	In	Out	Owner
Solar Panels	Power batteries through sunlight	Solar Panels	Battery Charging	Power Subsystem
Attitude Sensors	Determine orientation of the satellite	ACDS Computer	ACDS Computer	ACDS Subsystem
Deployment Mechanisms	Deploy the solar array	Main Computer	Deployment Mechanisms	Structures Subsystem Caden

Interface Specification Table

From Main Computer:

Interface	Format	Protocol	Error Handling	Examples
ADCS Computer	packet	CAN BUS	CRC Check Bit stuffing Frame Check Retry	12 bytes { CAN STR: 0x00 CAN ID: 0x00 CAN Payload 0x00000000 CAN CRC: 0x00 CAN ETX: 0x00 }
Power Computer	Packet	CAN BUS	CRC Check Bit stuffing Frame Check Retry	12 bytes { CAN STR: 0x00 CAN ID: 0x00 CAN Payload 0x00000000 CAN CRC: 0x00 CAN ETX: 0x00 }
Image Adapter	Packet	CAN BUS	CRC Check Bit stuffing Frame Check Retry	12 bytes { CAN STR: 0x00 CAN ID: 0x00 CAN Payload 0x00000000 CAN CRC: 0x00 CAN ETX: 0x00 }
Deployment Mechanisms	Signal(H/L)	GPIO	Retry	0 is High to Low 1 is Low to High {PIN ID:X DATA:0} {PIN ID:X DATA:1}
Secondary Payload	Frame	UART	Parity Error CRC Check ACK/NACK Retry	10+N byte (fixed) { Preamble: 0x00 NodeID: 0x00 FramID: 0x00 Payload: dynamic 0x0... Checksum: 0x00 ETX: 0x00 }
Telemetry / Command Radio	Frame	UART	Parity Error CRC Check ACK/NACK Retry	10+N byte (fixed) { Preamble: 0x00 NodeID: 0x00 FramID: 0x00 Payload: dynamic 0x0... Checksum: 0x00 ETX: 0x00 }

Data Storage	I/O	SPI	Retry Device Reset	N/A
Data Processor	Packet	CAN BUS	CRC Check Bit stuffing Frame Check Retry	CAN ID: 0x10 CAN Payload 0x00000000

From Deployment Mechanisms:

Interface	Format	Protocol	Error Handling	Examples
Main Computer	Signal(H/L)	GPIO	Retry	0 is High to Low 1 is Low to High {PIN ID:X DATA:0} {PIN ID:X DATA:1}

From Attitude Sensors:

Interface	Format	Protocol	Error Handling	Examples
ADCS Computer	Frame	I2C	N/A (handled by subsystem team)	N/A (handled by subsystem team)

From Attitude Control Actuators:

Interface	Format	Protocol	Error Handling	Examples
ADCS Computer	Frame	I2C	N/A (handled by subsystem team)	N/A (handled by subsystem team)

From GPS Receiver:

Interface	Format	Protocol	Error Handling	Examples
ADCS Computer	I/O	SPI	N/A (handled by subsystem team)	N/A (handled by subsystem team)
GPS Antenna	I/O	SPI	N/A (handled by hardware)	N/A (handled by hardware)

From ADCS Computer:

Interface	Format	Protocol	Error Handling	Examples
Attitude Sensor	Frame	I2C	N/A (handled by subsystem team)	N/A (handled by subsystem team)
Attitude Control Actuator	Frame	I2C	N/A (handled by subsystem team)	N/A (handled by subsystem team)
GPS Receiver	I/O	SPI	N/A (handled by subsystem team)	N/A (handled by subsystem team)
Main Computer	Packet	CAN BUS	CRC Check Bit stuffing Frame Check Retry	12 bytes { CAN STR: 0x00 CAN ID: 0x00 CAN Payload 0x00000000 CAN CRC: 0x00 CAN ETX: 0x00 }

From Secondary Payload:

Interface	Format	Protocol	Error Handling	Examples
Main Computer	Frame	UART	Parity Error CRC Check ACK/NACK Retry	10+N byte (fixed) { Preamble: 0x00 NodeID: 0x00 FramID: 0x00 Payload: dynamic 0x0... Checksum: 0x00 ETX: 0x00 }

From Imager Adapter:

Interface	Format	Protocol	Error Handling	Examples
Main Computer	Frame	CAN BUS	CRC Check Bit stuffing Frame Check Retry	12 bytes { CAN STR: 0x00 CAN ID: 0x00 CAN Payload 0x00000000 CAN CRC: 0x00 CAN ETX: 0x00 }
Truth Imager	Direct	Camera Link	N/A (handled by subsystem team)	N/A (handled by subsystem team)
Multispectral Imager	Direct	Camera Link	N/A (handled by subsystem team)	N/A (handled by subsystem team)
Data Processor	Frame	UART	Parity Error CRC Check ACK/NACK Retry	N byte (dynamic) { Preamble: 0x00 ID: 0x00 Length1: 0x00 Length2: 0x00 Payload: dynamic 0x0... Checksum: 0x00 }

From Truth Imager:

Interface	Format	Protocol	Error Handling	Examples
Imager Adapter	Direct	Camera Link	N/A (handled by subsystem team)	N/A (handled by subsystem team)

From Multispectral Imager:

Interface	Format	Protocol	Error Handling	Examples
Imager Adapter	Direct	Camera Link	N/A (handled by subsystem team)	N/A (handled by subsystem team)

From Battery Charging:

Interface	Format	Protocol	Error Handling	Examples
Power Computer	I/O	SPI	N/A (handled by subsystem team)	N/A (handled by subsystem team)

From Battery Monitor

Interface	Format	Protocol	Error Handling	Examples
Power Computer	I/O	SPI	N/A (handled by subsystem team)	N/A (handled by subsystem team)

From Power Computer

Interface	Format	Protocol	Error Handling	Examples
Battery Charging	I/O	SPI	N/A (handled by subsystem team)	N/A (handled by subsystem team)
Battery Monitor	I/O	SPI	N/A (handled by subsystem team)	N/A (handled by subsystem team)
Main Computer	Packet	CAN BUS	CRC Check Bit stuffing Frame Check Retry	12 bytes { CAN STR: 0x00 CAN ID: 0x00 CAN Payload 0x00000000 CAN CRC: 0x00 CAN ETX: 0x00 }

From Data Storage:

Interface	Format	Protocol	Error Handling	Examples
Main Computer	I/O	SPI	Retry Device Reset	{ SS: 0x00 SCLK: 0x00 MOSI: 0x00 MISO: 0x00 }
Data Processor	I/O	SPI	Retry Device Reset	{ SS: 0x00 SCLK: 0x00 MOSI: 0x00 MISO: 0x00 }

From Data Processor:

Interface	Format	Protocol	Error Handling	Examples
Main Computer	Packet	CAN BUS	CRC Check Bit stuffing Frame Check Retry	CAN ID: 0x10 CAN Payload 0x00000000
Data Storage	I/O	SPI	Retry Device Reset	{ SS: 0x00 SCLK: 0x00 MOSI: 0x00 MISO: 0x00 }
Imager Adapter	Frame	UART	Parity Error CRC Check ACK/NACK Retry	N byte (dynamic) { Preamble: 0x00 ID: 0x00 Length1: 0x00 Length2: 0x00 Payload: dynamic 0x0... Checksum: 0x00 }
Main Downlink Radio	I/O	SPI	Retry Device Reset	{ SS: 0x00 SCLK: 0x00 MOSI: 0x00 MISO: 0x00 }

From Telemetry/ Command Radio

Interface	Format	Protocol	Error Handling	Examples
Main Computer	Frame	UART	Parity Error CRC Check ACK/NACK Retry	10+N byte (fixed) { Preamble: 0x00 NodeID: 0x00 FramID: 0x00 Payload: dynamic 0x0... Checksum: 0x00 ETX: 0x00 }
Power Computer	Frame	UART	Parity Error CRC Check ACK/NACK Retry	10+N byte (fixed) { Preamble: 0x00 NodeID: 0x00 FramID: 0x00 Payload: dynamic 0x0... Checksum: 0x00 ETX: 0x00 }
UHF Antenna	Direct	Direct	N/A (handled by hardware)	N/A (handled by hardware)

From Main Downlink Radio

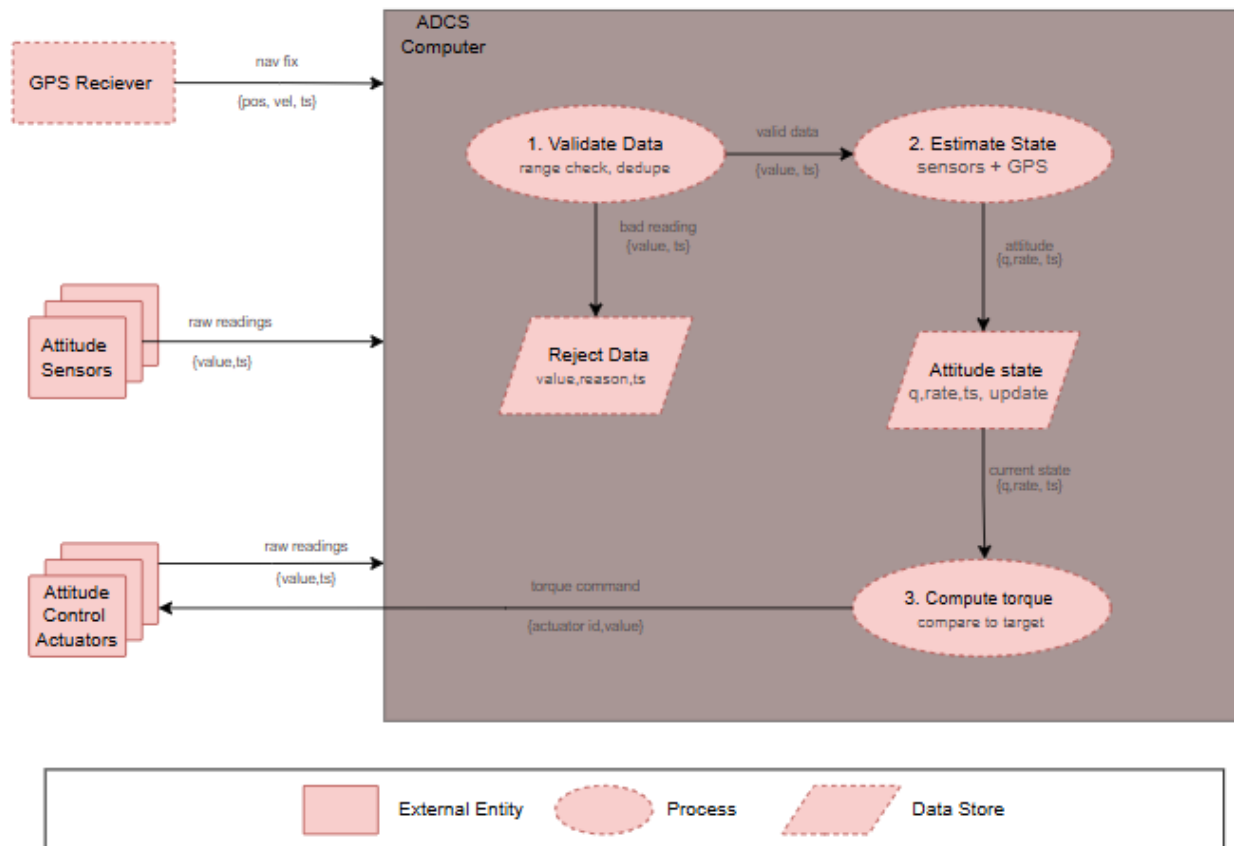
Interface	Format	Protocol	Error Handling	Examples
Data Processor	I/O	SPI	Device Reset	{ SS: 0x00 SCLK: 0x00 MOSI: 0x00 MISO: 0x00 }
S-band Antenna	Direct	Direct	N/A (handled by hardware)	N/A (handled by hardware)

Data Flow Diagrams

ADCS Data Flow

ADCS Data Flow

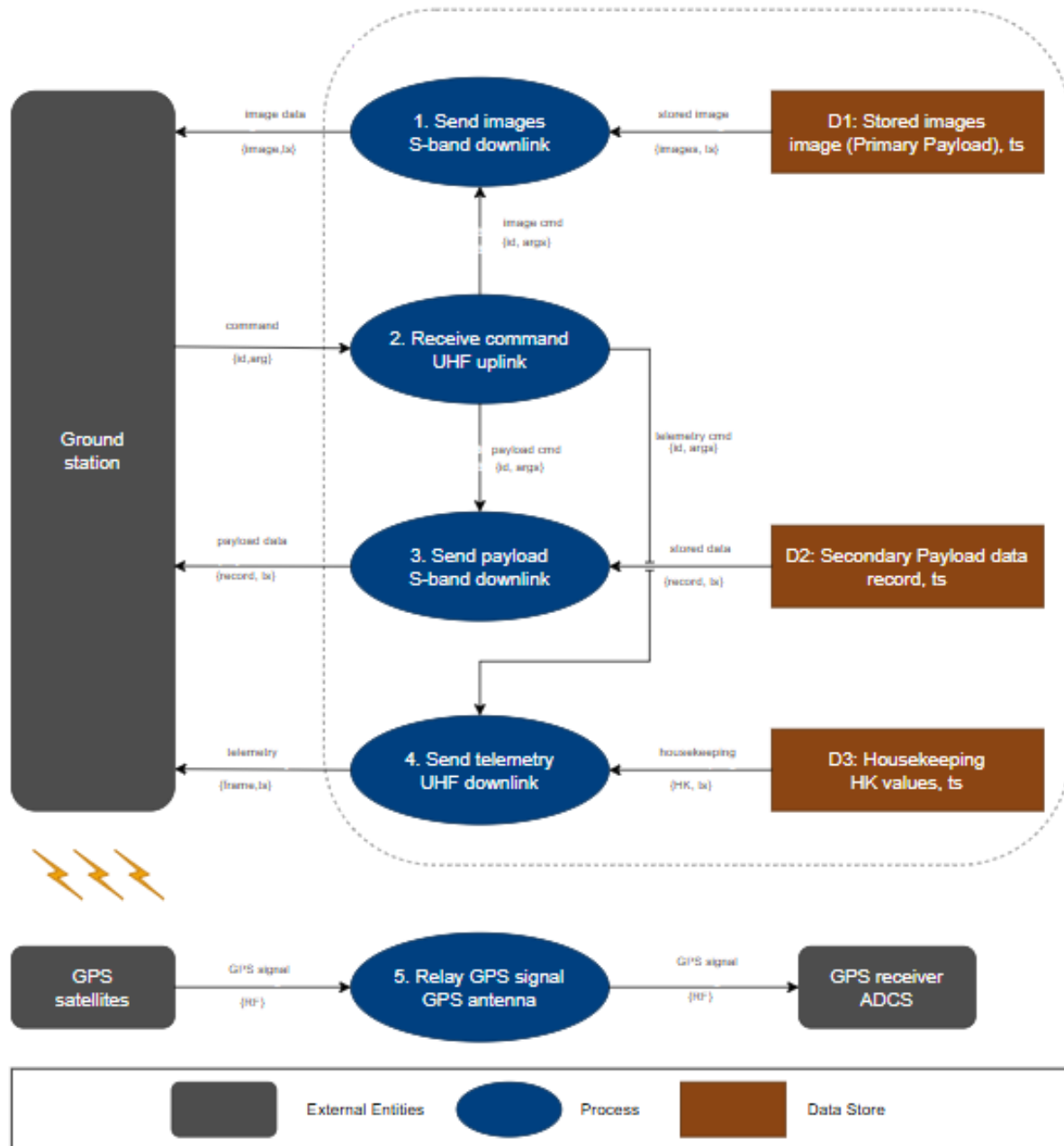
Shows how the ADCS computer takes an array of inputs and determines how to actuate to orient the satellite in the desired manner.



Communications Data Flow

Communications Data Flow

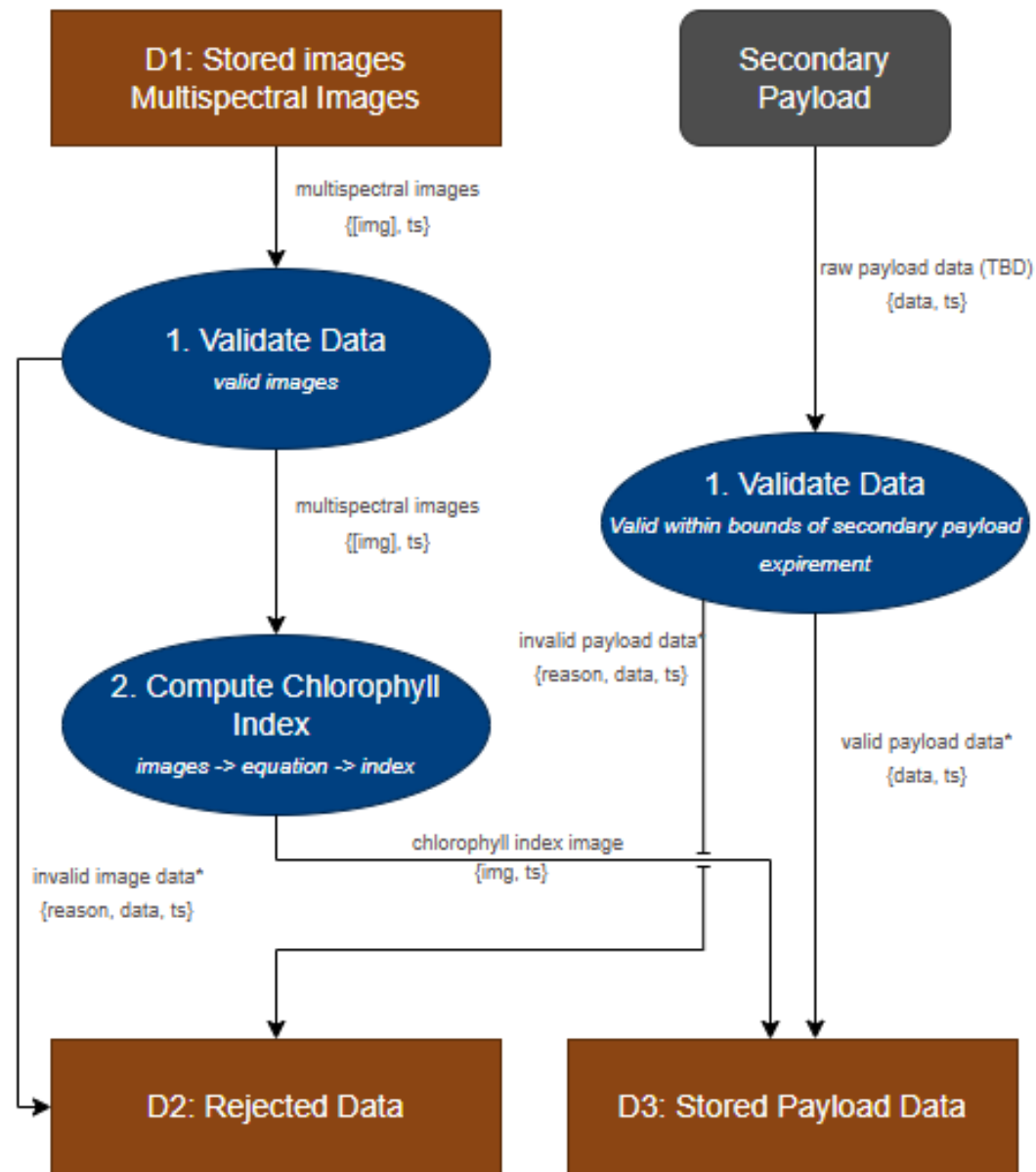
Shows how the satellite communicates with external systems to extract and provide mission data.



Payload Data Flow

Payload Data Flow

Shows how payload data is transferred to storage and processed into a meaningful index.

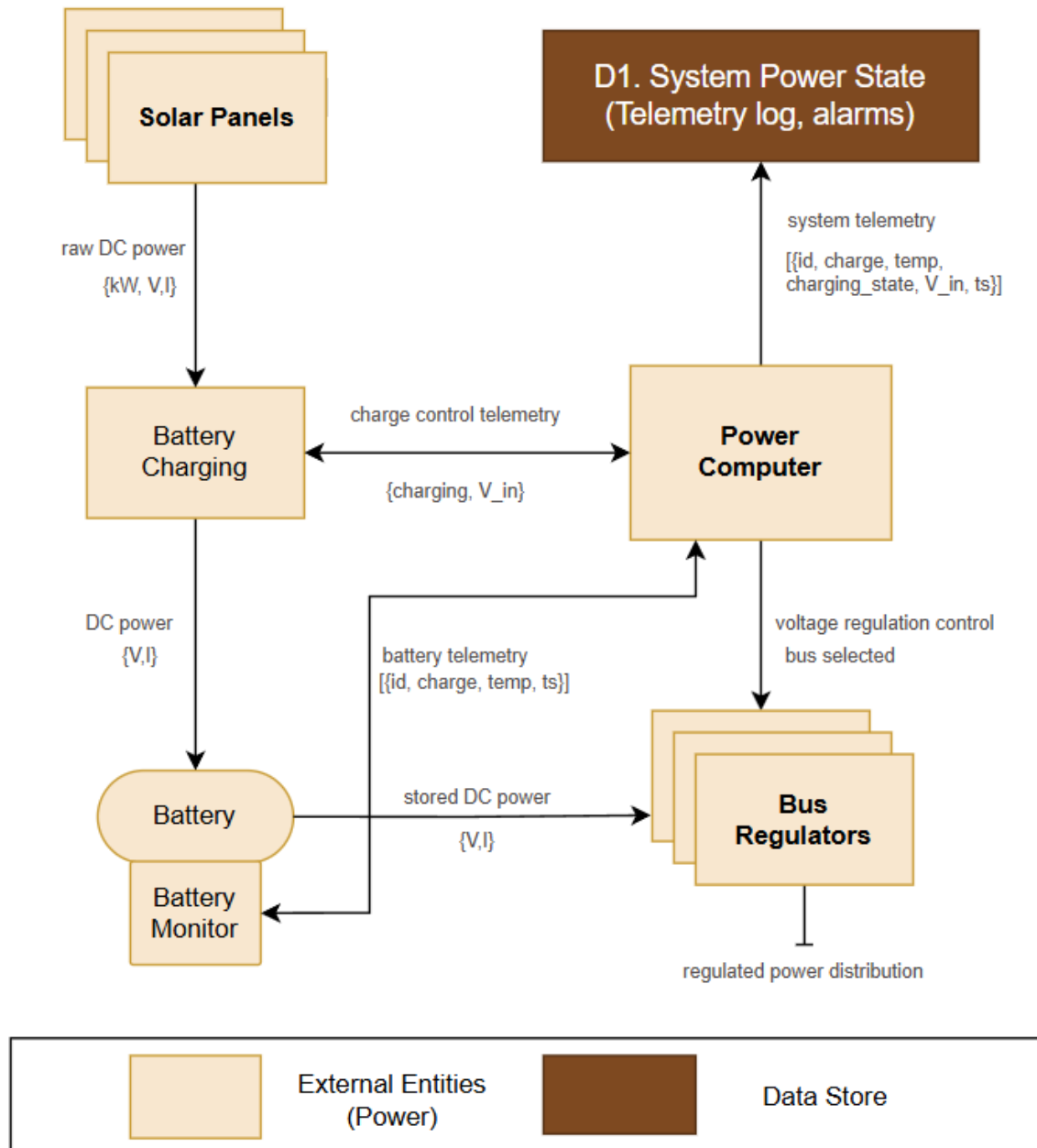


*Secondary payload is still not defined for the mission yet, so more specific information is TBD.



Power Data Flow

Shows how the satellite communicates with external power systems charge, and record battery health



Architecture Pattern

For the HABSat-1 On-Board Computer (OBC) flight software, we use a combination of the Embedded Sensor-Actuator pattern, complemented by a Client-Server model for ground station links.

Embedded Sensor-Actuator Pattern

Governs the low-level hardware interaction layers, managing telemetry acquisition and actuator commands across satellite subsystems (e.g., EPS, ADCS, payloads).

Client-Server Pattern

Governs the communication interface and command/telemetry exchange between the OBC and the ground station.

Fit to problem shape

The data and control in a satellite move hierarchically and cyclically—subsystem telemetry flows upward, CDH application logic processes commands, and control signals flow back down to actuators. The **Embedded Sensor-Actuator** and **Layered** patterns directly match this unidirectional and hierarchical flow of control and data between hardware devices and mission tasks.

Team skillset

As a student-led team (CubeCats), the architecture must be realistic to build, debug, and maintain. Layered and embedded sensor-actuator designs are standard paradigms taught in embedded systems courses, ensuring the team has the practical competency to implement them without getting bogged down by overly complex, esoteric frameworks.

Performance & timing needs

Spacecraft flight software must meet strict real-time deadlines and execute scheduled mission tasks deterministically. A **Layered** design ensures clear, bounded execution pathways with minimal dynamic overhead, keeping latency predictable and preventing priority inversions during critical flight states.

Scalability needs

HABSat-1 may need to accommodate future mission phases, new payload data, and additional telemetry channels. The **Layered** pattern handles this scalability gracefully by allowing developers to plug in new application modules or expand CDH capabilities at the top layer without rewriting lower-level hardware drivers.

Hardware constraints

CubeSat On-Board Computers operate under tight hardware limits (low processing power, constrained memory, and strict power budgets). This pattern combination avoids high memory footprints, dynamic allocations, and the processing overhead of distributed architecture, fitting cleanly onto embedded microcontrollers or single-board computers.

Rejected Pattern

Microservices

Reason for Rejection

Microservices were rejected because they introduce excessive network overhead, inter-process communication complexity, and memory footprint requirements. Running independent microservices across containerized or networked nodes exceeds the constrained processing and resource limits of a single-board CubeSat OBC.

Decision Log

Architecture Decision Records (ADRs), covering the project design choices for HABSat-1:

ADR-001: Addition of a Dedicated Data Processor

- **Title:** ADR-001: Addition of a Dedicated Data Processor
- **Status:** Accepted, Week 5 (9/30/2026). Revisit if payload data throughput requirements decrease.
- **Context:** High-bandwidth payload tasks (such as image capture and processing) threaten to saturate the CPU and memory resources of the primary On-Board Computer (OBC).
- **Options considered:** A. Process all data and telemetry on the single main OBC, B. Add a dedicated secondary data processor/coprocessor, C. Offload all raw processing completely to the ground station.
- **Decision:** Option B. Offloads compute-heavy payload tasks, protecting core Command and Data Handling (CDH) reliability and real-time scheduling.
- **Consequences:** Adds hardware mass, power drawing, and inter-processor communication complexity.
- **Owner and reviewers:** CubeCats OBC Team; reviewed by senior chief engineer on September 30th.

ADR-002: Route S-Band Downlink Connection Directly to Data Processor

- **Title:** ADR-002: Route S-band downlink connection to data processor
- **Status:** Accepted, Week 5 (9/30/2026). Revisit if primary OBC bandwidth bottlenecks disappear.
- **Context:** High-rate image files and science data compiled by the newly added data processor need efficient transmission paths to the S-band radio without bottlenecking the main OBC.
- **Options considered:** A. Route all payload files through the main compute before downlink, B. Connect the S-band downlink directly to the data processor, C. Store data entirely on external flash until ground command retrieval.
- **Decision:** Option B. Bypasses main computer bottlenecks for high-rate data transmission, maximizing downlink efficiency.

- **Consequences:** Requires multiplexing or arbitration logic to manage radio access between the main OBC (telemetry/commands) and the data processor (science data).
- **Owner and reviewers:** CubeCats OBC and Comms Team; reviewed by senior chief engineer on September 30th.

ADR-003: Transition Image Adapter Connection from UART to CAN Bus

- **Title:** ADR-003: Transition image adapter connection from UART to CAN
- **Status:** Accepted, Week 5 (9/30/2026). Revisit if data rate requirements exceed CAN bus limits.
- **Context:** The current point-to-point UART connection linking the image adapter lacks robust multi-drop capabilities, error-checking, and noise immunity for reliable satellite data bus route.
- **Options considered:**
 - A. Keep point-to-point UART.
 - B. Upgrade to high-speed SPI.
 - C. Transition to a Controller Area Network (CAN) bus.
- **Decision:** Option C. Provides robust multi-drop capabilities, superior error handling, and high noise immunity suited for harsh space environments.
- **Consequences:** CAN controller hardware support on endpoints and implementation of higher-layer bus protocol handling.
- **Owner and reviewers:** CubeCats OBC Team; reviewed by senior chief engineer on September 30th.

ADR-004: Attach Backup Telemetry Radio Connection to Power Computer

- **Title:** ADR-004: Attach telemetry radio connection to power computer for failover
- **Status:** Accepted, Week 5 (9/30/2026). Revisit if main computer reliability testing proves fault-proof.
- **Context:** A catastrophic failure of the main computer would result in total loss of communication and mission awareness unless a failsafe reporting mechanism exists.
- **Options considered:**

- A. Rely solely on main computer watchdog resets.
 - B. Install a completely independent secondary watchdog board.
 - C. Attach a backup connection from the Power Computer (EPS) to the telemetry radio.
- **Decision:** Option C. Enables critical health beacon transmissions during main computer faults using existing power subsystem hardware.
- **Consequences:** Introduces a cross-subsystem hardware dependency between power and communications; requires failover switching protocol logic.
- **Owner and reviewers:** CubeCats OBC and Power Teams; reviewed by senior chief engineer on September 30th.